

Gauss' Law for Central Forces with Inverse Square Characteristic

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August 2026

Introduction

A central problem in physics was to show that the gravitational field due to a sphere was the same as that produced by a point of corresponding mass. The result is indeed true; however, one can derive it beautifully using Gauss' Law. This paper first derives Gauss' Law for the electric and gravitational fields, which motivates the proof of its general form for central forces of the form $k/r^2 \hat{r}$.

The paper then finishes with a derivation of the gravitational field due to a sphere using the machinery developed in the previous sections.

As always, comments on pedagogical style and suggestions are most welcome:

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1 The Electric Field

Recall that by Coulomb's Law, the electric field due to a point charge q centered at the origin is given by

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

where r denotes the distance from the charge.

Consider a sphere S of radius r around this charge, as shown in the figure below.

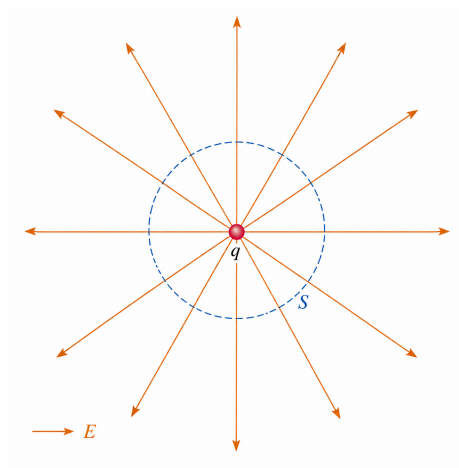


Figure 1: *The sphere S which encloses the charge q , where the origin is taken at. Image generated by ChatGPT 5.5.*

The *electric flux*¹ passing through the sphere is given by the surface integral

$$\Phi = \oint_S \vec{E} \cdot d\vec{a}. \quad (1)$$

Now, the area element is given by $d\vec{a} = r^2 \sin \theta d\theta d\phi \hat{r}$

Substituting our expressions for the electric field and $d\vec{a}$ into our integral above yields

¹You can think of the electric flux as a measure of the number of field lines passing through the sphere.

$$\Phi = \oint_S \vec{E} \cdot d\vec{a} \quad (2)$$

$$= \oint_S \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \cdot r^2 \sin\theta \, d\theta \, d\phi \, \hat{r} \quad (3)$$

$$= \frac{q}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\pi \sin\theta \, d\theta. \quad (4)$$

Now, we *could* evaluate the surface integral by brute force, as below:

$$\begin{aligned} \int_0^{2\pi} \int_0^\pi \sin\theta \, d\theta \, d\phi &= \int_0^{2\pi} [-\cos\theta]_0^\pi \, d\phi \\ &= 2 \int_0^{2\pi} d\phi \\ &= 4\pi. \end{aligned}$$

However, if we pay more attention to the integrand: $\sin\theta \, d\theta \, d\phi$, we would notice that it is simply the area element for the unit sphere: $da_{S^2} = \sin\theta \, d\theta \, d\phi$

In particular, Equation 4 transforms into

$$\begin{aligned} \Phi &= \frac{q}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\pi da_{S^2} \\ &= \frac{q}{\epsilon_0}. \end{aligned}$$

Hence, we have derived the equation

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q}{\epsilon_0}.$$

But there's an issue: The equation above is only for a spherical surface! However, if you consider an arbitrary surface S' that encloses the S (to which we know the flux), we will notice that the number of field lines crossing S' is the same as that for S , so that the fluxes are the same ³

$$\Phi_{S'} = \oint_{S'} \vec{E} \cdot d\vec{a} = \Phi_S = \frac{q}{\epsilon_0}.$$

²The S^2 is not related to the original sphere S ; rather, it is just mathematical notation denoting the unit sphere in $\mathbb{R}^3$.

³I am simplifying here; you can refer to page 27 of [Jac98] if you feel uncomfortable with this line of logic.

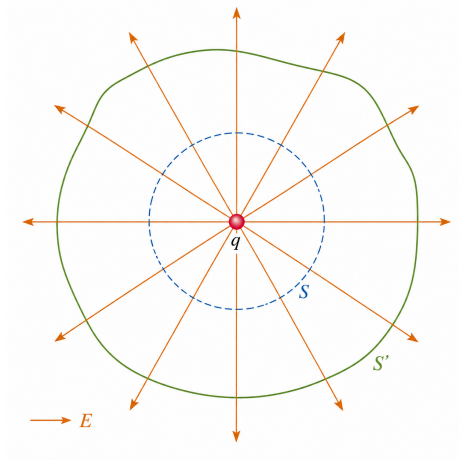


Figure 2: *Field lines passing through S and S' . Image generated by ChatGPT 5.5.*

Evidently, if we have an array of charges $q_1, \dots, q_n$ which are enclosed by a surface S , then since

$$\vec{E} = \sum_{k=1}^N \vec{E}_k,$$

the flux through S is given by⁴

$$\begin{aligned} \oint \vec{E} \cdot d\vec{a} &= \oint \sum_{k=1}^n \vec{E}_k \cdot d\vec{a} \\ &= \sum_{k=1}^n \oint \vec{E}_k \cdot d\vec{a} \\ &= \sum_{k=1}^n \frac{q_k}{\varepsilon_0} \\ &= \frac{1}{\varepsilon_0} \sum_{k=1}^n q_k \\ &= \frac{1}{\varepsilon_0} Q_{\text{enc}}. \end{aligned}$$

⁴It should be noted that charges *outside* the surface will *not* contribute to the total flux, since the field lines pass through the surface twice— the dot products of the field with an area element will cancel.

I will explain why this result (and the analogous one for gravitational fields in Section 2) is important at the end of the paper.

2 The Gravitational Field

After reading the first section, you might ask the question: "Does something similar also hold for gravitational fields?"

This is a good line of intuition since the electric and gravitational fields due to a point mass and charge situated at the origin, respectively, are given by⁵

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}, \quad \vec{g} = \frac{GM}{r^2} \hat{r}$$

It is, therefore, reasonable to make the following conjecture, known as Gauss' Law for gravity.

Proposition. *Consider a closed surface S . Then we have*

$$\oint_S \vec{g} \cdot d\vec{a} = 4\pi GM_{\text{enc}}.$$

A quick remark: Suppose you were a physicist deriving these equations when they were unknown— you wouldn't know the true relation beforehand, so you would have to guess what the flux of the gravitational field through a surface would be equal to, and then prove it. But how can we make a reasonable guess? Here's something to consider: remember from Section 1 that our surface integral produced a factor of 4π , which was eventually multiplied by the Coulomb constant $\frac{1}{4\pi\epsilon_0}$. This suggests that if our intuition is correct (in the sense that the forces can be treated in essentially the same way), the constant Gm will be multiplied by a factor of 4π after the dust has settled in evaluating the integral. And in fact, it is correct.

Proof: Again, we consider a point mass M situated at the origin, enclosed in a sphere S of radius r . The flux through S is given by the surface integral

$$\oint_S \vec{g} \cdot d\vec{a}$$

We apply the same strategy for the gravitational field, substituting the surface area element $d\vec{a} = r^2 \sin \theta d\theta d\phi \hat{r}$ into the equation above:

⁵Intuitively speaking, they are the 'same' force— just multiplied by a constant

$$\begin{aligned}
\oint_S \vec{g} \cdot d\vec{a} &= \int_0^{2\pi} \int_0^\pi \frac{Gm}{r^2} \vec{r} \cdot (r^2 \sin \theta d\theta d\phi \hat{r}) \\
&= Gm \int_0^{2\pi} \int_0^\pi \sin \theta d\theta d\phi \\
&= 4\pi GM.
\end{aligned}$$

The same argument as outlined in Section 1 for the electric field shows that this holds for an arbitrary surface S' , that masses *outside* do not contribute to the total flux, and therefore

$$\oint_{S'} \vec{g} \cdot d\vec{a} = 4\pi GM_{\text{enc}},$$

which completes the proof. ■

3 Generalisation to Central Forces

The previous two sections have hinted toward a more general result:

Generalised Gauss' Law.

Suppose a particle with some physical quantity β situated at the origin induces a field (which obeys the principle of superposition) $\vec{K}$ of the form

$$\vec{K} = \frac{k\beta}{r^2} \hat{r}$$

where k is a constant independent of r .

Then, for any closed surface S ,

$$\oint_S \vec{K} \cdot d\vec{a} = 4\pi k\beta_{\text{enc}}$$

Proof: Enclose the particle in a sphere S , of radius r . Then

$$\begin{aligned}
\oint_S \vec{K} \cdot d\vec{a} &= \int_0^{2\pi} \int_0^\pi \left(\frac{k\beta}{r^2} \hat{r} \right) \cdot (r^2 \sin \theta d\theta d\phi \hat{r}) \\
&= k\beta \int_0^{2\pi} \int_0^\pi \sin \theta d\theta d\phi \\
&= 4\pi k\beta.
\end{aligned}$$

The flux through the sphere will be the same as the flux through an arbitrary surface S enclosing the particle, so that

$$\oint_{\mathcal{S}} \vec{K} \cdot d\vec{a} = 4\pi k\beta.$$

Note that particles outside $\mathcal{S}$ will not contribute to the overall field.

Now, consider an array of particles $\beta_1, \dots, \beta_n$. These particles will induce fields $\vec{K}_1, \dots, \vec{K}_n$. Our superposition assumption implies that the overall field is given by the sum $\vec{K} = \vec{K}_1 + \dots + \vec{K}_n$. Hence,

$$\begin{aligned} \oint_{\mathcal{S}} \vec{K} \cdot d\vec{a} &= \oint_{\mathcal{S}} \sum_{j=1}^n \vec{K}_j \cdot d\vec{a} \\ &= \sum_{j=1}^n \oint_{\mathcal{S}} \vec{K}_j \cdot d\vec{a} \\ &= \sum_{j=1}^n 4\pi k\beta_j \\ &= 4\pi k\beta_{\text{enc}}, \end{aligned}$$

which proves the theorem. ■

4 Significance of Gauss' Law

We return to Gauss' Law for the gravitational field to conclude the paper. Note that the formula

$$\vec{g} = \frac{GM}{r^2} \hat{r}$$

only gives the gravitational field due to a *point* mass. Obviously, in the real world, while planets can be treated as point particles at large distances, this approximation quickly fails as one approaches a planet's surface. We need an understanding of the gravitational field just around a planet if we want to understand the orbits of satellites. A sphere is a reasonable approximation for most planets, such as Earth. We therefore need a way to calculate the gravitational field produced by a sphere of mass M . A sphere is not a point mass. Luckily, Gauss' Law saves us from performing difficult calculations:

Consider a sphere of mass M and radius R centered at the origin, and suppose we are at a distance $r \geq R$ from the center of the sphere⁶. Let the gravitational field due to this sphere be denoted by $\vec{g}$. By Gauss' Law in Section 2, we have

⁶We must make this assumption to ensure $M_{\text{enc}} = M$.

$$\oint \vec{g} \cdot d\vec{a} = 4\pi GM. \quad (5)$$

Here, our area element is given by $d\vec{a} = r^2 \sin \theta d\theta d\phi \hat{r} = da \hat{r}$. Certainly, the gravitational field will point radially outward from the sphere's surface, i.e., $\vec{g} = g \hat{r}$, so evaluating the scalar product:

$$\begin{aligned} \oint \vec{g} \cdot d\vec{a} &= g \oint da \\ &= 4\pi gr^2. \end{aligned}$$

So by Equation 5:

$$4\pi gr^2 = 4\pi GM$$

which implies

$$g = \frac{GM}{r^2}$$

and, in particular

$$\vec{g} = \frac{GM}{r^2} \hat{r}$$

which shows that the gravitational field outside a sphere is the same as if the field were produced by a point mass!

References

- [Jac98] John David Jackson. *Classical Electrodynamics*. 3rd ed. New York: John Wiley & Sons, 1998. ISBN: 0-471-30932-X.